UChicago REU Notes Week 1

Zih-Yu Hsieh

June 21, 2026

Contents

1 Week 1

Here it's mainly the preparation for any further theory in spectra. To learn the material, here I focused mainly on some topological preliminaries. Main reference comes from *Concise Course in Algebraic Topology*.

Remark 1.1. I believe most texts assume that compactness means not only the existence of finite subcover for arbitrary open cover, but also Hausdorffness on the space. Here, we also assume this.

1.1 Compactly Generated Spaces

In most literatures I've read, the spaces are assumed to be compactly generated. So, this section is collecting some information I have about it. First, starting with the notion of k-spaces:

Definition 1.2. Given a topological space X, a subset $A \subseteq X$ is *k-closed*, if for any compact space K, and continuous map $g : K \to X$, the preimage $g^{-1}(A) \subseteq K$ is closed.

X is a *k-space*, if any k-closed subset $A \subseteq X$ is closed.

Which, adapting the usual continuous maps, this realizes a *Category of k-spaces*, denoting $k\mathcal{U}$. There are some properties about this category, regarding a functor from regular category of spaces $\mathcal{U}$, and this category's completeness / cocompleteness.

Definition 1.3. Given a space X, define a topological space kX where the underlying set coincides with X, while requiring all k-closed sets $A \subseteq X$ to be closed in kX as its new topology.

This construction satisfies the closed sets axioms, hence form a topology; also, under the new topology, kX is a k-space. Furthermore, since all closed sets are automaticlaly k-closed, the set identity function $\iota_X : kX \to X$ is in fact continuous.

We want to further argue k is a functor $\mathcal{U} \to k\mathcal{U}$, which can be achieved with the following proposition:

Proposition 1.4. *Let Z be a k-space, Y arbitrary topological space, and $f : Z \to Y$ a continuous map. Then, there exists a unique continuous map $kf : Z \to kY$, such that the following diagram commutes:*

$$\begin{tikzcd} Z \arrow[rr, "\forall f"] \arrow[dr, dashed, "\exists! kf"'] & & Y \\ & kY \arrow[ur, "\iota_Y"'] & \end{tikzcd}$$

Proof. Since ι_Y is a set identity map, uniqueness of kf is clear (as set function it coincides with f). To show kf is continuous, it suffices to show all k-closed set $A \subseteq Y$ has preimage $f^{-1}(A) \subseteq Z$ being closed.

Let $g : K \to Z$ be arbitrary continuous map, with K being compact. Consider $f \circ g : K \to Z \to Y$, for any $A \subseteq Y$ that's k-closed, $(f \circ g)^{-1}(A) = g^{-1}(f^{-1}(A)) \subseteq K$ is closed, hence $f^{-1}(A) \subseteq Z$ is also k-closed. Since Z is a k-space, $f^{-1}(A)$ is closed. $\square$

Corollary 1.5. *Given any continuous map $f : X \to Y$, there exists a unique continuous map $kf : kX \to kY$, such that the following diagram commutes:*

$$\begin{tikzcd} X \arrow[r, "\forall f"] & Y \\ kX \arrow[u, "\iota_X"] \arrow[r, dashed, "\exists! kf"'] & kY \arrow[u, "\iota_Y"'] \end{tikzcd}$$

Proof. Since $f \circ \iota_X : kX \to Y$ is continuous, with kX being a k-space, **Proposition 1.4** guarantees the existence and uniqueness of kf (which as set map it also agrees with f, as ι_X is set identity). $\square$

This formulates $k : \mathcal{U} \to k\mathcal{U}$ as a functor. And, if consider the inclusion functor $\mathcal{I} : k\mathcal{U} \to \mathcal{U}$, **Proposition 1.4** also insists $(\mathcal{I}, k)$ as an adjoint pair, as it phrases a natural isomorphism as follow:

$$\mathrm{Hom}_{\mathcal{U}}(\mathcal{I}Z, Y) \cong \mathrm{Hom}_{k\mathcal{U}}(Z, kY), \quad f \mapsto kf$$

Using some categorical nonsense, k as a right adjoint preserves all limits, so $k\mathcal{U}$ is a complete category. In paticular, we denote $X \times^k Y := k(X \times Y)$ as the product of two k-spaces X, Y.

To show it's also cocomplete, consider the following:

Proposition 1.6. *Quotient of a k-space is still a k-space; arbitrary disjoint union of k-spaces is a k-space.*

Proof. Let X be a k-space, and $f : X \to Y$ be a quotient map. For any k-closed subset $A \subseteq Y$, consider arbitrary continuous map $g : K \to X$ with K compact. Given the composition $f \circ g : K \to X \to Y$, $(f \circ g)^{-1}(A) = g^{-1}(f^{-1}(A)) \subseteq K$ is closed, hence $f^{-1}(A) \subseteq X$ is k-closed, which is closed as X is a k-space. Then, using the quotient space property, $A = f(f^{-1}(A)) \subseteq Y$ is closed, concluding that Y is also a k-space.

Now, given $\{X_i\}_i$ a family of k-spaces, consider $Z := \coprod_i X_i$, with $\iota_i : X_i \to Z$ the canonical inclusion. Given any k-closed subset $A \subseteq Z$, similar logic from previous proofs show that $A \cap X_i = \iota_i^{-1}(A) \subseteq X_i$ are all k-closed, hence closed (as X_i's are k-spaces). Which, this verifies $A \subseteq Z$ is closed under the disjoint union topology, showing Z is also a k-space. $\square$

This guarantees the corproduct and coequalizers of k-spaces (and maps between them) in $\mathcal{U}$ are still k-spaces, which these colimits exist in $k\mathcal{U}$, showing the cocompleteness.

Now, move on to another criterion for compactly generated spaces:

Definition 1.7. A space X is *k-Hausdorff* if the diagonal $\Delta X \subset k(X \times X)$ is closed.

Denote $h\mathcal{U}$ as the *Category of k-Hausdorff spaces.*

There's a lot more details for this, I'll try and fill it in afterward==.

There is another functor $h : \mathcal{U} \to h\mathcal{U}$ which generates a k-Hausdorff space out of any topological spaces, and with inclusion functor $\mathcal{J} : h\mathcal{U} \to \mathcal{U}$, they form an adjoint pair $(h, \mathcal{J})$. So, h as a left adjoint preserves colimits.

Finally, for compactly generated spaces, it's defined as a k-Hausdorff k-space. In this space, limits and colimits exist, but need to be constructed in $\mathcal{U}$, then apply the functor h or k to yield the desired results.

Another equivalent way to phrase k-Hausdorffness when given a k-space, is the *weak Hausdorffness*: Given any continuous map $g : K \to X$ with K compact, $g(K) \subseteq X$ is a closed subset.

1.2 Spaces of Maps

This section, we'd like to realize $Y^X := \mathrm{Hom}_{\mathsf{Top}}(X, Y)$ (the set of all continuous maps $X \to Y$) as a topological space. Here, we consider the *compact open topology.*

First, given the any map $f : K \times X \to Y$, there is a corresponding set function $\tilde{f} : K \to Y^X$ by $\tilde{f}(k) = f(k, \_) : X \to Y$.

Definition 1.8. The *compact open topology* on $\mathrm{Hom}_{\mathsf{Top}}(X, Y)$, has $S \subseteq Y^X$ being open iff all compact space K, continuous map $f : K \times X \to Y$, the set

$$\tilde{f}^{-1}(S) = \{k \in K | f(k, \_) \in S\}$$

is open in K.

Obviously, $\emptyset, Y^X$ are both open; if $\{S_i \subseteq Y^X\}_i$ are open, then $\tilde{f}^{-1}(\bigcup_i S_i) = \bigcup_i \tilde{f}^{-1}(S_i)$ is open in K, so $\bigcup_i S_i$ is open; finally, if $S, S' \subseteq Y^x$ are open, then $\tilde{f}^{-1}(S \cap S') = \tilde{f}^{-1}(S) \cap \tilde{f}^{-1}(S')$ is open in K, so $S \cap S'$ is also open. This defines a topology.

Here is some nice properties of composition of maps:

Proposition 1.9. *Given $g : X' \to X$, the precomposition $g^* := \_ \circ g : Y^X \to Y^{X'}$ is continuous. Dually, given $h : Y \to Y'$, the postcomposition $h_* := h \circ \_ : Y^X \to Y'^X$ is continuous.*

Proof. In the first scenario, given arbitrary open subset $S' \subseteq Y^{X'}$, the preimage $S := (g^*)^{-1}(S') = \{\ell \in Y^X | \ell \circ g \in S'\}$. Fix arbitrary continuous map $f : K \times X \to Y$ with K compact, consider $\tilde{f}^{-1}(S) \subseteq K$, we get:

$$\tilde{f}^{-1}(S) = \{k \in K | f(k, \_) \in S\} = \{k \in K | f(k, \_) \circ g \in S'\}$$

Similarly, consider $f' := f \circ (\mathrm{id}_K \times g) : K \times X' \to Y$ (by $f'(k, x') = f(k, g(x'))$), the set $\tilde{f'}^{-1}(S') \subseteq K$ is open by assumption, which it has the following description:

$$\tilde{f'}^{-1}(S') = \{k \in K | f'(k, \_) \in S'\} = \{k \in K | f(k, \_) \circ g \in S'\} = \tilde{f}^{-1}(S)$$

This shows the openness of $S \subseteq Y^X$.

For the second scenario, given arbitrary open subset $S' \subseteq Y'^X$, the preimage $S := h_*^{-1}(S') = \{\ell \in Y^X | h \circ \ell \in S'\}$. Fix arbitrary continuous map $f : K \times X \to Y$ with K compact, notice the following description:

$$\tilde{f}^{-1}(S) = \{k \in K | f(k, \_) \in S\} = \{k \in K | h \circ f(k, \_) \in S'\}$$

Which, notice $h \circ f : K \times X \to Y \to Y'$ satisfies $(h \tilde{\circ} f)^{-1}(S') = \{k \in K | h \circ f(k, \_) \in S'\} = \tilde{f}^{-1}(S)$, and by assumption this set is open. So, $S \subseteq Y^X$ is open, finishing the claim. $\square$

As a result, the Hom functor $\mathrm{Hom}_{\mathsf{Top}} : \mathsf{Top}^{\mathrm{op}} \times \mathsf{Top} \to \mathsf{Top}$ is a bifunctor.

If restrict to the compactly generated spaces, there is a homeomorphism

$$\mathrm{Hom}_{\mathsf{Top}}(X \times^k Y, Z) \cong \mathrm{Hom}_{\mathsf{Top}}(X, \mathrm{Hom}_{\mathsf{Top}}(Y, Z))$$

Yeah, more detailes to fill in.

1.3 Cofibration

Intuitively, these are the maps where homotopy can be extended to the codomain:

Definition 1.10. A map $\iota : A \to X$ is a *cofibration*, if it satisfies the *Homotopy ExtensionProperty* (HEP). This means, given any map $f : X \to Y$, homotopy $H : A \times I \to Y$, zeroth inclusions $i_0 : A \to A \times I$ and $i_0 : X \to X \times I$ (by $x \mapsto (x, 0)$), and product map $\iota \times \mathrm{id}_I : A \times I \to X \times I$, the homotopy extends to $\tilde{H} : X \times I \to Y$, with the following commutes:

$$\begin{tikzcd}
A \arrow[rr, "i_0"] \arrow[dd, "\iota"'] & & A \times I \arrow[dl, "H"'] \arrow[dd, "\iota \times \mathrm{id}_I"] \\
& Y & \\
X \arrow[ur, "f"] \arrow[rr, "i_0"'] & & X \times I \arrow[ul, dashed, "\exists \tilde{H}"']
\end{tikzcd}$$

Intuitively, it gives a description of how to map X into Y (also how to map A into Y), generates a homotopy H on A only, and the property guarantees it to extend to a homotopy on X.

It's preserved under fibre product:

Lemma 1.11. *If $\iota : A \to X$ is a cofibration, $g : A \to B$ any map, then the induced pushout $\iota' : B \to B \sqcup_A X$ is a cofibration:*

$$\begin{tikzcd}
A \arrow[r, "\iota"] \arrow[d, "g"'] & X \arrow[d, "g'"] \\
B \arrow[r, "\iota'"'] & B \sqcup_A X
\end{tikzcd}$$

Proof. Before starting, notice that product preserves adjunction/fibre coproduct, because $(B \sqcup_A X) \times I \cong (B \times I) \sqcup_(A \times I)(X \times I)$ (i.e. the first space can be viewed as the disjoint union of the product, then glue using identical methods at each $t \in I$).

Then, given map $f : B \sqcup_A X \to Y$, homotopy $H : B \times I$, and the fibre coproduct maps, we get:

$$\begin{tikzcd}
A \arrow[rrrr, "i_0"] \arrow[dr, "g"] \arrow[dddd, "\iota"'] & & & & A\times I \arrow[dl, "g\times \mathrm{id}_I"'] \arrow[dddd, "\iota\times\mathrm{id}_I"] \\
& B \arrow[rr, "i_0"] \arrow[dd, "\iota'"'] & & B\times I \arrow[dl, "H"'] \arrow[dd, "\iota'\times\mathrm{id}_I"] & \\
& & Y & & \\
& B\sqcup_A X \arrow[ur, "f"] \arrow[rr, "i_0"'] & & (B\sqcup_A X)\times I & \\
X \arrow[rrrr, "i_0"'] \arrow[ur, "g'"] & & & & X\times I \arrow[ul, "g'\times\mathrm{id}_I"'] \arrow[uull, dashed, bend left, "\exists\tilde{H}"]
\end{tikzcd}$$

The existence of $\tilde{H}$ is given by HEP of $\iota : A \to X$. This induces the following diagram:

$$\begin{tikzcd}
A\times I \arrow[r, "\iota\times\mathrm{id}_I"] \arrow[d, "g\times\mathrm{id}_I"'] & X\times I \arrow[d, "g'\times\mathrm{id}_I"] \arrow[ddr, bend left, "\tilde{H}"] & \\
B\times I \arrow[r, "\iota'\times\mathrm{id}_I"'] \arrow[drr, bend right, "H"'] & (B\sqcup_A X)\times I \arrow[dr, dashed, "\exists!\overline{H}"] & \\
& & Y
\end{tikzcd}$$

where $\overline{H}$ is induced by the fibre coproduct property. Plugin to the previous diagram, we yield the desired homotopy extension for the inner square. $\square$

If consider the "special case" $Y = M_\iota$, the mapping cylinder of $\iota : A \to X$ (which is constructed as $M_\iota := (A \times I \sqcup X)/\sim$, where $(a, 0) \sim f(a)$). Notice it generates the following fibre coproduct diagram:

$$\begin{tikzcd}
A \arrow[r, "i_0"] \arrow[d, "\iota"'] & A\times I \arrow[d, "H"] \\
X \arrow[r, "i_0'"'] & M_\iota
\end{tikzcd}$$

So, the HEP and fibre coproduct property guarantees the following diagram:

$$\begin{tikzcd}
A \arrow[rr, "i_0"] \arrow[dd, "\iota"'] & & A\times I \arrow[dl, "H"'] \arrow[dd, "\iota\times\mathrm{id}_I"] \\
& M_\iota \arrow[dr, dashed, "\exists!j"'] & \\
X \arrow[ur, "i_0'"] \arrow[rr, "i_0"'] & & X\times I \arrow[ul, dashed, bend right, "\exists\tilde{H}"']
\end{tikzcd}$$

Here, $\tilde{H}$ is the homotopy extension, j is the factorization by fibre coproduct. And, this enforces $\tilde{H} \circ j = \mathrm{id}_{M_\iota}$, showing j is injective. If restrict to $t = 1$, then $j : A \times \{1\} \to X \times \{1\}$ is precisely the map ι, showing it's injective (in fact it's an embedding, because we also have a retract $\tilde{H} : X \times \{1\} \to A \times \{1\}$).

From other sources I can find, ι unfortunately doesn't have closed image in general, yet for the case of A, X being compactly generated (for instance, CW Complexes) I believe this is true. (Maybe by k-closed property).

So, we see $\iota : A \to X$ is a cofibration (for compactly generated spaces) directly implies ι is an embedding with closed image (or A is a closed subspace of X); and, the mapping cylinder is precisely $M_\iota = X \times \{0\} \cup A \times I$ in $X \times I$, the extended homotopy guarantees it's a retract of $X \times I$.

Conversely, given $A \subseteq X$ a closed subspace, while its mapping cylinder $M_\iota \subseteq X \times I$ is a retract of $X \times I$. Then, use the retract as the extended homotopy (together with the fact that it's a fibre coproduct, hence the "initial extension diagram" whenever seeking the homotopy extension for other Y), we get $A \hookrightarrow X$ is a cofibration.

So, for compactly generated spaces, it's equivalent to say: a cofibration is a closed embedding $A \subseteq X$, such that the mapping cylinder $X \times \{0\} \cup A \times I$ is a retract of $X \times I$.

To talk about homotopy equivalences of cofibers (which are not necessarily cofibrations, but just maps $A \to X$ when fixing A). Given, $i : A \to X$, $j : A \to Y$ two cofibers, a map $f : X \to Y$ is a morphism, if the following diagram commutes:

$$\begin{tikzcd} & A \arrow[dl, "i"'] \arrow[dr, "j"] & \\ X \arrow[rr, "f"'] & & Y \end{tikzcd}$$

Definition 1.12. Two such morphisms $f, f' : X \to Y$ are homotopic in this category (under A), if they're homotopic relative to A, i.e. there exists homotopy $H : X \times I \to Y$ of f, f', where all $a \in A, t \in I$ satisfies $H(i(a), t) = j(a)$ (namely A is fixed under this homotopy). This is called *cofiber homotopy.*

The homotopy equivalence built on cofiber homotopies is called *cofiber homotopy equivalence.*

This notion of homotopy equivalence is much stronger than original homotopy. Yet, when restricting to cofibrations, then it doesn't make a difference:

Proposition 1.13. *Given $i : A \to X$, $j : A \to Y$ two cofibrations, if morphism $f : X \to Y$ is a homotopy equivalence, then it's a cofiber homotopy equivalence.*

Proof. It suffices to find $g : Y \to X$ that's a homotopy invese, with $g \circ f \simeq \mathrm{id}_X$ relative to A (because then apply the same argument on $g : Y \to X$, there is a homotopy inverse $f' : X \to Y$, such that $f' \circ g \simeq \mathrm{id}_Y$ relative to A. As a result, $f = \mathrm{id}_Y \circ f \simeq f' \circ g \circ f \simeq f' \circ \mathrm{id}_X = f'$ relative to A, hence $f \circ g \simeq f' \circ g \simeq id_Y$ relative to A, showing it's a cofiber homotopy equivalence).

First, by hypothesis, there is $g'' : Y \to X$ a homotopy inverse of f. (Not done) □

1.4 Fibration

As a dual notion to cofibration (which "extends" homotopy to the target space), fibration is the one that "lifts" the homotopy to the source space.

Definition 1.14. A surjective map $p : E \twoheadrightarrow B$ is a *fibration*, if it satisfies *covering homotopy property* (CHP in May's book), or *homotopy lifting property.* Meaning, given any $f : Y \to E$, and homotopy $H : Y \times I \to B$

of $p \circ f : Y \to E \twoheadrightarrow B$, there exists a homotopy $\tilde{H} : Y \times I \to E$ of f, with the following diagram comutes:

$$\begin{tikzcd}
Y \arrow[r, "\forall f"] \arrow[d, "i_0"'] & E \arrow[d, two heads, "p"] \\
Y \times I \arrow[r, "\forall H"'] \arrow[ur, dashed, "\exists \tilde{H}"] & B
\end{tikzcd}$$

A typical example is the covering map, where homotopies can be lifted uniquely once a basepoint of the covering is chosen. So, fibration can be viewed as a generalization of covering space, where homotopy liftings exist.

Using the k-ified compact-open topology of the function space (in particular, the homeomorphism $\mathrm{Hom}_{\mathsf{Top}}(K \times X, Y) \cong \mathrm{Hom}_{\mathsf{Top}}(K, \mathrm{Hom}_{\mathsf{Top}}(X, Y))$, and $Y^X := \mathrm{Hom}_{\mathsf{Top}}(X, Y)$), the above can be reformulated as the following (which is structure wise more similar to the dual of the cofibration diagram):

$$\begin{tikzcd}
E \arrow[dd, "p"'] & & E^I \arrow[ll, "p_0"'] \arrow[dd, "p \circ -"] \\
& Y \arrow[ul, "\forall f"] \arrow[ur, dashed, "\exists \tilde{H}"] \arrow[dr, "\forall H"'] & \\
B & & B^I \arrow[ll, "p_0"]
\end{tikzcd}$$

Then, by dualizing the proof for cofibration, we yield the following result:

Lemma 1.15. *Pullback of fibration is fibration: Given fibration $p : E \twoheadrightarrow B$, and any map $g : B' \to B$, the base change $p' : E \times_B B' \to B'$ is a fibration:*

$$\begin{tikzcd}
E \times_B B' \arrow[r, "g'"] \arrow[d, two heads, "p'"'] & E \arrow[d, two heads, "p"] \\
B' \arrow[r, "g"'] & B
\end{tikzcd}$$

Similar to the existence of mapping cylinder M_ι as the "initial test diagram" for cofibration (as it's the fibre coproduct of cofibration $A \to X$, and inclusion $A \to A \times I$), it suggests the exitence of "final test diagram" for fibration, when choosing $Y = N_p := E \times_B B^I$ (the fibre product of fibration $E \to B$, and evaluation by zero map $B^I \to B$). The fibre product property (together with the test diagram of fibration) guarantees N_p is the final diagram (namely, all other choices of Y factors uniquely through N_p).

A precise description of N_p is as follow:

$$N_p = \{(e, \beta) | \beta(0) = p(e)\} \subseteq E \times B^I$$

Plugin to the diagram, we observe the following using the fibration and fibre product property:

$$\begin{tikzcd}
E \arrow[dd, "p"'] & & E^I \arrow[ll, "p_0"'] \arrow[dd, "p \circ -"] \arrow[dl, dashed, "\exists ! k"'] \\
& N_p \arrow[ul, "p_1"] \arrow[ur, dashed, "\exists s"'] \arrow[dr, "p_2"'] & \\
B & & B^I \arrow[ll, "p_0"]
\end{tikzcd}$$

Here, k is induced by fibre coproduct, s is induced by fibration property. Universality of N_p (as fibre product) guarantees that $k \circ s = id_{N_p}$, another name for these lifting s is called *path lifting function* (as it lifts a pair $(e, \beta) \in N_p$ with $\beta(0) = p(e)$, to a path $\tilde{beta} \in E^I$, that satisfies $\tilde{\beta}(0) = e$, and $p \circ \tilde{beta} = \beta \in B^I$).

A special case - when $p : E \to B$ is a covering, then s is in fact unique, due to the unique path lifting property (whenever fixing a path and a basepoint).

For common types of fibrations, the case for covering space suggests the notion of fibre bundle could work: Given covering $p : E \to B$, every point $x \in B$ has an open neighborhood $U_x \subseteq B$, such that $p^{-1}(U_x) \cong U_x \times p^{-1}(x)$ (where $p^{-1}(x)$ is a discrete subspace), and all fibres are homeomorphic to each other (i.e. same cardinality when the fibre is discrete).

If restricts the bundle to have *numerable open covers* (namely, an open cover $\{U_i\}_i$, such that each point $x \in B$ is contained in finitely many U_i's, and each U_i is equipped with a continuous function $\lambda_i : B \to I$ such that $\lambda_i^{-1}(0, 1] = U_i$). Then, the following theorem holds:

Theorem 1.16. *Given a map $p : E \to B$ and $\{U_i\}_i$ a numerable open cover of B. Then, p is a fibration iff the restrictions $p|_{U_i} : p^{-1}(U_i) \to U_i$ is a fibration for every U_i.*

This guarantees all fibre bundles with numerable trivialization cover to be a fibration.

Similar to cofibre homotopy, we define fibre homotopy as follow:

Definition 1.17. Given two fibres $p : D \to B$ and $p' : E \to B$ (Note: they're not necessarily fibrations), a map $f : D \to E$ is a morphism if the following diagram commutes:

$$\begin{tikzcd} D \arrow[rr, "f"] \arrow[dr, "p"'] & & E \arrow[dl, "p'"] \\ & B & \end{tikzcd}$$

And, given two morphisms $f, f' : D \to E$, they're *fibre homotopic* if there is a homotopy $H : D \times I \to E$, such that $p = p' \circ H : D \to B$ (namely, at all time $t \in I$, the map preserves p).

By dualizing the notion fro cofibre, we again have the following criterion of fibre homotopy equivalence:

Proposition 1.18. *Let $p : D \to B$, $q : E \to B$ be two fibrations. If $f : D \to E$ be a homotopy equivalence morphism (i.e. $q \circ f = p$), then it's a fibre homotopy equivalence.*

To talk about the relations between the fibres (for instance, for fibre bundle all fibres are homeomorphic), the use of paths (as homotopy between points) can also b lifted given fibration, and can be used to study such relation.

Let $p : E \to B$ be a $\beta : I \to B$ be a path with $\beta(0) = b$ and $\beta(1) = b'$, and $F_x := p^{-1}(x) \subseteq E$ as the fibre (as subspace under E). Then, one can consider the following diagram, using the lifting property:

$$\begin{tikzcd} F_b \times \{0\} \cong F_b \arrow[rr, "\iota_b"] \arrow[d, "i_0"'] & & E \arrow[d, "p"] \\ F_b \times I \arrow[r, "\pi_2"'] \arrow[urr, dashed, "\exists \tilde{\beta}"] & I \arrow[r, "\beta"'] & B \end{tikzcd}$$

Which, specify to $t = 1$ for $\tilde{\beta}$, we realize the following:

$$p \circ \tilde{\beta}_1(F_b) = \beta \circ \pi_2(F_b \times \{1\}) = \beta(1) = b' \implies \tilde{\beta}_1(F_b) \subseteq F_{b'}$$

So, it induces a map $\tilde{\beta}_1 : F_b \to F_{b'}$ (more generally, it induces a continuous family of maps $\tilde{\beta}_t : F_b \to F_{\beta(t)}$.

Proposition 1.19. *Suppose paths $\beta, \beta' : I \to B$ are homotopic, then the induced maps $\tilde{\beta}_1, \tilde{\beta'}_1 : F_b \to F_{b'}$ are also homotopic.*

Proof. Let $H : I \times I \to B$ be the path homotopy of $\beta \simeq \beta'$, so that $H(s,0) = \beta(s)$, $H(s,1) = \beta'(s)$, $H(0,t) = b$, and $H(1,t) = b'$. If consider the subspace of $I \times I$, say $J := (I \times \{0\}) \cup (I \times \{1\}) \cup (\{0\} \times I)$ (in the homotopy H, first side represents β, second side represents β', and the third side represents the side that's constant as b), notice there is an automorphism on $I \times I$, so that J gets mapped to $I \times \{0\}$, so the lifting property can be used whenever a map is defined on $J \subset I \times I$.

Now, define the map map $f : F_b \times J \to E$, so that on $I \times \{0\}$ it's $\tilde{\beta} : F_b \times I \cong F_b \times I \times \{0\} \to E$, on $I \times \{1\}$ it's $\tilde{\beta}' : F_b \times I \cong F_b \times I \times \{1\} \to E$, and on $\{0\} \times I$ it's the map $\iota_b \circ p_1 : F_b \times I \cong F_b \times \{0\} \times I \to F_b \to E$ (more precisely, we only consider the value in F_b, and include it into E). By the defining property of each $\tilde{\beta}, \tilde{\beta}'$, these three maps glue together, hence f is well-defined.

Then, consider the inclusion $i : J \hookrightarrow I \times I$, we form the following diagram using the lifting property:

$$\begin{tikzcd}
F_b \times J \arrow[rr, "f"] \arrow[d, "\mathrm{id}_{F_b} \times i"'] & & E \arrow[d, "p"] \\
F_b \times I \times I \arrow[r, "\pi_{2,3}"'] \arrow[urr, dashed, "\exists \tilde{H}"] & I \times I \arrow[r, "H"'] & B
\end{tikzcd}$$

Which, if restrict to $F_b \times I \times \{0\}$, $\tilde{H}$ restricts to $\tilde{\beta}$ by our chohice of f; similarly, restricting to $F_b \times I \times \{1\}$, $\tilde{H}$ restricts to $\tilde{\beta}'$, showing the two homotopic. So, specify to $t = 1$, we get $\tilde{\beta}_1 \simeq \tilde{\beta'}_1$, showing the map $[\tilde{\beta}_1] : F_b \to F_{b'}$ is a well-defined map between fibres, and dependent only on homotopy classes of paths $[\beta] : I \to B$, connectinb b to b'. $\square$

Corollary 1.20. *Given a fibration, any two points in the same path component have homotopy equivalent fibres.*

Proof. This is because if $\beta, \gamma : I \to B$ are two paths such that $\beta(1) = \gamma(0)$, then if $b := \beta(0), b' := \beta(1) = \gamma(0), b'' := \gamma(1)$, then $\tilde{\beta}_1 : F_b \to F_{b'}$ and $\tilde{\gamma} : F_{b'} \to F_{b''}$ composes to be the same as $(\gamma \tilde{\cdot} \beta)_1 : F_b \to F_{b''}$ (it's induced by the concatination of maps). So, $[\gamma \tilde{\cdot} \beta_1] = [\tilde{\gamma}_1] \circ [\tilde{\beta}_1]$ when passing to homotopy classes of maps.

Then, since a constant path c_b incudes identity $[\tilde{c}_{b1}] : F_b \to F_b$, we see $[\widetilde{\beta^{-1}}_1] \circ [\tilde{\beta}_1] = [\widetilde{\beta^{-1}\dot{\beta}_1}] = [\tilde{c_p}] = [id_{F_b}]$, showing $[\tilde{\beta}_1]$ is a homotopy equivalence. $\square$

1.5 Based Homotopy

Definition 1.21. Given a based topological space X, its *reduced/based suspension* $\Sigma X := X \times I / X \vee I$, where $X \vee I$ is the wedge sum of X and inteveral I once fixed a basepoint $x_0 \in X$ and $i \in I$, while $X \times I$ has basepoint $(x_0, i) \in X \times I$.

Equivalently, $\Sigma X \cong X \wedge S^1 := X \times S^1 / X \vee S^1$.